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Forecasting of Groundwater Level in Multiple Wells using Machine Learning Models

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***Abstract*— Accurate forecasting of groundwater levels is crucial for sustainable water resource management and environmental planning. This article explores the use of machine learning for accurate groundwater level predictions. In this study three machine learning models including artificial neural network ANN, long-short term memory LSTM and Extreme Gradient Boosting decision tree XGBoost technique were used to predict ground water level. We incorporate deep water surface measurements and meteorological data as key parameters to enhance forecasting accuracy.**

Keywords: Groundwater Level, ANN, LSTM, XGBoost, time series forecasting

I. INTRODUCTION

Groundwater, as one of the world's most valuable resources, plays an important role in sustaining the environment, supporting agriculture, and meeting the growing need for drinking water. However, sustainable management of soil resources is becoming a complex challenge in the face of climate change, population growth and environmental regulations. Timely and accurate forecasting of groundwater levels is a key component of effective groundwater management, enabling us to make informed decisions about water availability, conservation and protection [1][2].

In this context, this study presents an innovative and robust approach to groundwater level prediction, exploiting the formidable capabilities of Artificial Neural Networks (ANN), the Deep Learning Algorithm Long-Short Term Memory (LSTM)networks and the powerful XGBoost Algorithm. These techniques, grounded in the principles of time series forecasting, hold immense promise for unlocking valuable insights from historical groundwater data, enabling us to make more informed decisions for the future[19].

What distinguishes this methodology is its integration of deep water surface measurements, raising the accuracy of forecasts to new heights. The heart of this approach lies in its meticulous data preparation and modeling techniques. We start by preprocessing and merging deep water surface data, creating a comprehensive dataset of unparalleled richness. This dataset serves as the basis on which we build and refine our forecasting models, using LSTM, ANN and XGBoost.

The main objective of this study is to improve the accuracy and reliability of groundwater level forecasts to promote sustainable water resources management and environmental planning[20]. To achieve this ambitious goal, we have set the following specific goals:

• Develop a new method that uses the power of Artificial Neural Networks (ANN), Long-Short Term Memory (LSTM) and XGBoost Algorithms for groundwater level prediction.

• Include water surface measurements as a critical parameter to create a comprehensive dataset that captures both long-term trends and short-term changes in groundwater levels.
• Train and refine LSTM, ANN and XGBoost models using comprehensive data sets to improve prediction accuracy and robustness.

The reminder of this paper is organized into several key sections to systematically address the objectives and proposed solution. Section 2 carries a Literature Review where we delve into existing research and methodologies related to groundwater level prediction, highlighting gaps and the need for our innovative approach. Section 3 presents details on the collection and integration of deep water surface measurements, as well as the preprocessing steps presented in this section. Section 4 thoroughly explain the methodology used, including the architecture of LSTM and ANN, the implementation of the XGBoost algorithm, and the training process. Section 5 presents the results of our experiments, showing the increased accuracy and reliability of our models compared to traditional methods. We also discuss the implications of our findings. In Section 6, we summarize the main conclusions and contributions of our research, highlighting the importance of our approach in groundwater level prediction. In the Section 7 we list a comprehensive list of references that were included to support our research and methodology.

Through this well-structured organization, we aim to provide a comprehensive understanding of our research, from its creation and objectives to the implementation of the proposed solution and its implications for water resources management and environmental planning.

II. LITERATURE REVIEW

This literature review examines the use of Artificial Intelligence (AI), metaheuristics, the Internet of Things (IoT) and Machine Learning in groundwater water level prediction, highlighting examples of significant work:

A. Artificial Intelligence (AI)

Researchers have applied Deep Neural Networks to model water level fluctuations. For example, the Malik and Bhagwat [3] study used Artificial Neural Network to

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Model groundwater level fluctuations in urban areas where they predicted the pre and post monsoon groundwater fluctuation at the Tikri Kalan observation well situated in the West Delhi, India.

Furthermore, a large volume of literature is available on the application of ANN models to predict groundwater levels in different regions. Rajaee, Ebrahimi and Nourami [4] conducted a review of the Artificial Intelligence methods in groundwater level modeling that presents a brief overview of the most popular AI techniques, as well as bibliographic reviews of the authors' experiences over the years.

Nair and Sindhu [5] conducted a study to estimate the groundwater level in the Mamom river basin in the Trivandrum region of India. They used an ANN model based on hydrological parameters to estimate groundwater levels in three wells during the monsoon and monsoon seasons. Prediction models were built using only four weather factors as predictors: rainfall (Raf), evapotranspiration capacity (TEU), temperature (T) and humidity (H%).

A Feed Forward Artificial Neural Network (FFANN) was used by [6] to predict hourly groundwater level on eight wells in South Korea. The study took into account surface water level and groundwater basin parameters as input data. The estimated groundwater level values are very accurate, consistent with the actual groundwater level intensity, and thus the proposed model is considered effective in capturing.

B. Metaheuristics

Metaheuristics, such as Genetic Algorithms, have been used to optimize the parameters of prediction models. Kayhomayoon, Babaeiam and Milan [7] developed a Combination of Metaheuristic Optimization Algorithms and Machine Learning Methods Improves the Prediction of Groundwater Level where they improved the model and prediction performance using genetic algorithms such as particle swarm optimization (PSO), ant colony optimization for continuous domains (ACOR), and differential evolution (DE).

K.Roy, K.Biswas and A.Mattar [8] conducted a study about Groundwater level prediction using a multi-objective genetic algorithm-gray relational analysis based on a weighted set of ANFIS models. The aim of their study is to evaluate the prediction accuracy and estimation ability of different models based on the adaptive neuro-fuzzy inference system "ANFIS". These models include Differential Evolutionary ANFIS (DE-ANFIS), Particle Swarm Optimization-ANFIS (PSO-ANFIS) and Hybrid Algorithm Optimized Traditional ANFIS (HA-ANFIS) for forward forecasting groundwater levels for one or several weeks at three observation wells.

M.Adnan, Liang Dai and R.Mostafa [9] developed a model of groundwater level fluctuations by ELM merged advanced metaheuristic algorithms using hydroclimatic data. The study compares new extreme learning methods (ELM) adapted to metaheuristic algorithms such as particle swarm optimization (PSO), grey wolf optimization, whale optimization algorithm (WOA), l Harris Hawks Optimizer (HHO) and Jellyfish Search Optimizer (JFO) in groundwater level estimation.

C.Internet of Things (IoT)

The integration of IoT sensors in groundwater wells enables real-time monitoring, H F Hawari and P P Chantar [10] developed an IoT Real-Time groundwater monitoring system, using flow sensors to obtain data on groundwater consumption and send it to a server where all this data will be recorded for future analysis. They used Long Range (LoRa) communication module together with Thingspeak cloud and mobile application to test the developed system where the results showed promising coverage with Line of Sight (LOS) at 900m maximum while for Non-Line of Sight (NLOS) at 600m.

PAZHANI JAISUDHAN [11] suggested an IoT based groundwater level monitoring system for calculating and sensing water levels. The proposed technology is a wireless groundwater monitoring system. The prototype sensor module records and sends real-time groundwater level data from remote locations. It can be used for bore wells and agricultural purposes. IoT, a web and mobile monitoring service protocol, connects all data information.

Q. Zhou, C. Chen and G.Zhang highlighted in their literature review [12] the current groundwater management models with the IoT contributions where they reveal that groundwater managers do not have the effective, real-time groundwater management systems needed to collect the necessary data. Furthermore, the literature shows that current data collection methods do not have the efficiency needed to meet computational model requirements and achieve management goals. Unlike previous surveys that only focused on specific groundwater issues related to management methods through simulation and optimization.

D. Machine Learning (ML)

Many Machine-Learning techniques were used for forecasting groundwater level. Tao, M. Hameed and A. Marhoon [13] conducted a comprehensive review about groundwater level prediction using Machine Learning. The study aims to provide a clear understanding of the state-of-the-art ML models implemented for groundwater level (GWL) modeling and the milestones achieved in this field. The review covers all types of ML models used to model GWL from 2008 to 2020 about 138 articles and a detailed summary of the reviewed articles, including model type, data duration, time scale, input and output parameters, the used performance criteria and the best models are identified.

A.Hussein, Thron and Chaziasgar [14] proposed a machine learning approach for groundwater forecasting with three features: the use of regression-based approach to predict groundwater images based on a series of monthly groundwater maps, strategic automatic local and global feature selection using XGBoost, the use of multiple Machine Learning techniques such as XGBoost, Random Forest and Support Vector Regression. Among these techniques, support vector regression consistently performs best in terms of minimizing mean squared error and mean absolute error.

In summary, these research examples demonstrate how AI, metaheuristics, IoT, and Machine Learning are used in concert to improve groundwater level prediction. These approaches promote better management of groundwater resources, which is essential to meet the challenges of sustainable management of these resources.

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III. DATA COLLECTION

A. *Study area*

Water potential is already overexploited, especially by agriculture. This region is predominantly rural (63% of its population lives from working the land) and agriculture accounts for 90% of water consumption.

The numerous actors responsible for the exploitation and distribution of water find it very difficult to take coordinated action. Their less than rational management of Marrakech's water resources leads to the frequent water cuts experienced by the city and multiple conflicts of use between domestic needs, agricultural, industrial and tourist needs.

Water potential is already overexploited, especially by agriculture. This region is predominantly rural (63% of its population lives from working the land) and agriculture accounts for 90% of water consumption.

The Haouz water table has decreased by 20 meters in less than 30 years.

Tourism, which continues to grow, is the other major consumer. We now know that the city currently has, more than 30 golf courses, and will need 39 million m3 of water per year. The precarious balance, hitherto difficult to maintain, between water reserves and needs, risks being disrupted.

Collections in this region have decreased by 20% in recent years. If we add to this rainfall deficit the fact that existing resources have been abused, we arrive at a drop in underground reserves while we already know that the demand for drinking water will increase from 65 million m3 in 2011 to 78 million in 2015 and 114 million in 2030. Which means that in 2015, the city will experience a water deficit of 13 million m3 per year, which will worsen, reaching 52 million m3 in 2030.

In Marrakech, agriculture and tourism, which are extremely demanding of drinking water, will have to change their habits: groundwater reserves are decreasing and reservoirs are slowly emptying. There is no longer enough rain and underground water is clearly overexploited. In 2015, Marrakech will find itself with a water deficit of 13 million m3. The city is beginning to learn how to rationalize its resources, notably by embarking on the treatment and reuse of wastewater.

The Marrakech region has no choice, it must rationalize its water resources. Which will prove difficult in a region that is developing at a breakneck pace.

In this study, we examine the performance of groundwater level prediction in 4 groundwater wells in the Haouz area (Fig. 1). In this study, we exclusively consider daily time steps for water data. underground and meteorological data.

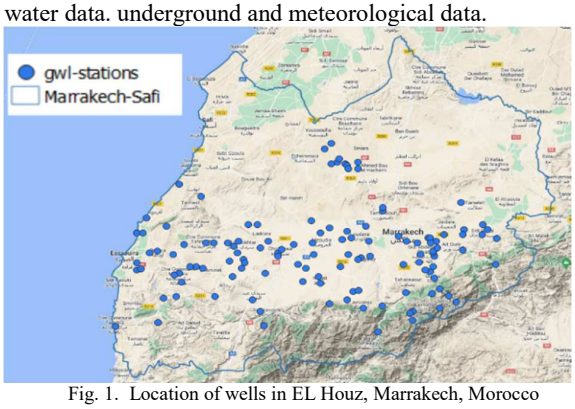


Fig. 1. Location of wells in EL Houz, Marrakech, Morocco

B. *Collected data description*

The groundwater data for this paper is from wells in EL Houz area in Marrakech, Morocco. Which contains different parameters used in this study such as reference point elevation, ground surface elevation, depth to the water surface, depth below ground surface and water surface elevation presented in Table 1.

Column	Type	Description
STATION	text	Unique Station Identifier (the state Well number)
MSMT_DATE	timestamp	Date/Time (in PST) the groundwater level measurement was collected
WLM_RPE	numeric	Reference Point Elevation used to collect the groundwater level measurement
WLM_RPE_QC	numeric	Quality Code for WLM_RPE measurement.
WLM_GSE	numeric	Ground Surface Elevation at the well site
WLM_GSE_QC	numeric	Quality Code for WLM_GSE measurement.
RPE_WSE	numeric	Depth to water surface in feet below reference point
RPE_WSE_QC	numeric	Quality Code for RPE_WSE measurement.
GSE_WSE	numeric	Depth below ground surface or the Distance from ground surface to water surface in feet
GSE_WSE_QC	numeric	Quality Code for GSE_WSE measurement.
WSE	numeric	Water Surface Elevation in feet above Mean Sea Level (NAVD88)
WSE_QC	numeric	Quality Code for WSE measurement.

Table .1. Dataset parameters and explanation

C. *Data preprocessing*

The preprocessing phase of the data is an essential step toward accurate groundwater level prediction. In this phase, the collected raw data contained errors, inconsistencies and messing values. To exploit the full potential of Machine Learning techniques, it is imperative to clean, normalize, and pre-process data meticulously. This involves resolving outliers, filling gaps in the data set, and ensuring data consistency. Proper preprocessing not only improves the quality of input data but also paves the way for more robust and reliable predictions, ultimately leading to more informed groundwater management decisions.

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IV. METHODOLOGY

A. *Artificial neural network (ANN)*

Generally, Neural Network (NN) models are used as very powerful Machine Learning Algorithms for predicting time series of various engineering applications. The ANN model consists of an input layer (x_t), one or more hidden layers, and an output layer (y). Each hidden layer has weight (w_t) and bias parameters (b) to manage the neurons. To transfer data from the hidden layer to the output layer, an activation function (a) is used[15][16]. With these parameters, the output can be represented as:

$$\boldsymbol{y = a(\sum_{i=1}^{n} x_i w_i + b)} \quad (1)$$

The feed forward neural network (FFNN) is a widely used ANN model. In this network, information moves only in the forward direction, from the input layer, through hidden layers, and finally to the output layer. In this case, the input layer is the number of neurons determined by the number of features; the output layer, the final layer in the NN with a single neuron; and the first hidden layer has 64 neurons, and 32 neurons in the second hidden layer. The FFNN model is developed by connecting the above neurons into one network (Fig.2).

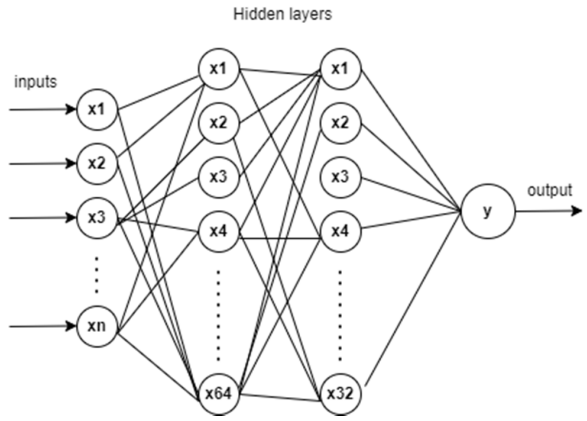


Fig. 3. Structure of the FFNN model

The activation function (a) used is ReLU.

ReLU Activation Function:

the Rectified Linear Unit, or ReLU, is a basic activation function. It's a straightforward but effective decision that improves the network's capacity to recognize and adjust to intricate patterns in the data.

ReLU is described mathematically as f(x)=max (0, x), which means that when the input is positive or zero, it returns the data exactly as it is, and when the input is negative, it becomes inactive and outputs zero. Because of its piecewise-linear behavior, which helps to mitigate the vanishing gradient problem, this study's groundwater level prediction is ideal for deep learning applications and is computationally efficient.

For the optimization algorithm used to compile this model is Adaptive Moment Estimation (Adam).

Adam Optimization Algorithm:

Adam, is a well-liked optimization method that is applied to neural network training. Combining aspects of RMSprop and momentum, it provides adaptable learning rates for increased productivity. Adam makes specific adjustments to learning rates for every model parameter, which makes the model ideal for deep learning tasks. This adaptability makes it an effective option for training neural networks in a variety of applications, including this study's groundwater level prediction, by accelerating convergence and assisting in overcoming obstacles like sparse gradients and noisy input.

B. *Long Short term memory (LSTM)*

Long Short-term memory networks are recurrent neural networks (RNN) widely applied to model sequential data such as time series or natural language. RNNs suffer from the problem of vanishing gradients during backpropagation, and in the case of simple RNNs, their memory barely covers the previous 10 steps. However, LSTM can remember long-term dependencies because they are explicitly designed to solve this problem. In addition to the hidden state of the RNNs, LSTMs have a cell memory to store information and three gates to control the information flow . The forget gate controls which and how much information of the cell memory is forgotten, the input gate controls which inputs are used to update the cell memory, and the output gate controls which elements of the cell memory are used to update the hidden state of the LSTM cell [16][17].

The LSTM cell with activation layers (input, output, forget gates, and the cell, connected with peephole connections) is shown in Fig. 4. Here, x_t is the input, f_t is the forget gate, it is the input gate, $\hat{c}_t$ is the cell update, C_t is the cell state, o_t is the output gate, and h_t is the output.

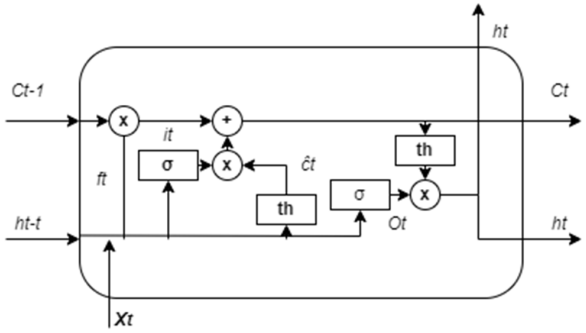


Fig. 4. Structure of the LSTM model

For this case the LSTM model has a layer with 62 units or neurons with a fully connected (Dense) layer to the model with 32 units and activation function set to ReLU. As for the compilation of the model, the Adam optimizer with a learning rate of 0.001 is used.

C. *Extrem gradient boosting (XGBoost)*

XGBoost is one of the implementations of gradient boosting machines, known as one of the highest performing algorithms used for supervised learning. It can be used for both regression and classification problems. Data scientists due to its high execution speed for basic computation prefer XGBoost [18].

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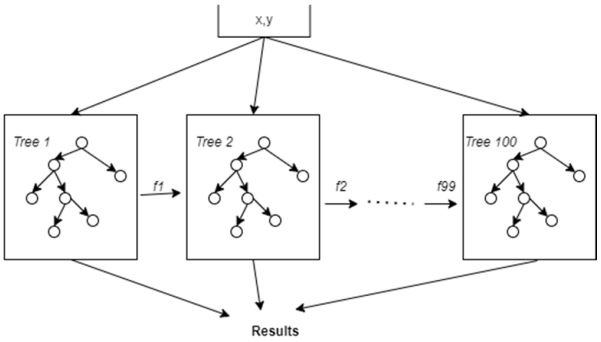


Fig. 5. Structure of the XGBoost model

Where x is the input, k represents the number of trees in the model, 100 in our case, f_k is the k-th tree and results is generated from the following equations:

$$\text{results} = \sum_{k=1}^{100} \text{f}_\text{k}(x) \quad (2)$$

The hyperparameters used here are set to 100 decision trees, a learning rate of 0.1 that means smaller steps and slower convergence, which can help prevent overfitting, and a maximum depth of 3 which means each tree can have a maximum of 3 levels of splits.

D. *Performance measures*

The performance of Machine Learning models needs to be evaluated. To do this, the results generated by the models are compared with the actual values of the groundwater level using the following performance indicators:

- Mean absolute error (MAE), which measures the how accurate the predicted values compared to the actual ones.

$$\mathbf{MAE} = \frac{1}{n}\sum_{i=0}^{n} |t_i - y_i| \quad (3)$$

- Root mean squared error (RMSE), measures the error between the model's output (y) and the target values (t).

$$\mathbf{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(t_i - y_i)2} \quad (4)$$

- Mean squared error (MSE), measures how well a regression fits the data.

$$\mathbf{MSE} = \frac{1}{n}\sum_{i=1}^{n}(t_i - y_i)2 \quad (5)$$

- R-Squared (R2), quantifies how well a regression model explains the variability in the target value.

$$\boldsymbol{R2} = 1 - \frac{SSR}{SST} \quad (6)$$

Where $\boldsymbol{n}$ is the number of data points, t_i is the observed values, y_i is the predicted values, $\boldsymbol{SSR}$ represents the sum squared differences between the observed values and the predicted values, and $\boldsymbol{SST}$ represents the sum of the squared differences between the observed values and the mean of the observed values.

V. RESULTS AND DISCUSSION

For validating the developed models, the dataset had been divided into training, validation and testing subsets represented in Table 2.

	ANN	LSTM	XGBoost
Testing	20%	15%	30%
Training	80%	70%	70%
Validation	--	15%	--

Table .2. Data percentage in testing, training and validation

A. *FFNN model*

The results of evaluating the FFNN model for predicting groundwater levels in four wells showed notable performance in the training set, demonstrating its ability to capture underlying patterns in the data. For Wells 3 and 4, as examples, results demonstrate strong training set performance, further highlighting the model's ability to fit the data accurately.

However, when tested on the validation set, the model performance, while still impressive, exhibited some well-to-well variation. For Well 1, the MSE was 0.24, the MAE was 0.06, and the R-squared was 0.79, indicating a relatively good fit but some room for improvement. In addition, Wells 3 and 4 also performed well on the test set, highlighting the generalization ability of the model.

These results show that the FFNN model is very effective in predicting groundwater levels, capable of providing accurate forecasts for most wells. Further adjustments and optimizations could potentially improve its performance, especially for wells with small deviations in the test set. Overall, these results highlight the promise of FFNN as a valuable groundwater-forecasting tool, providing valuable insights for practical applications and future research (Fig.6).

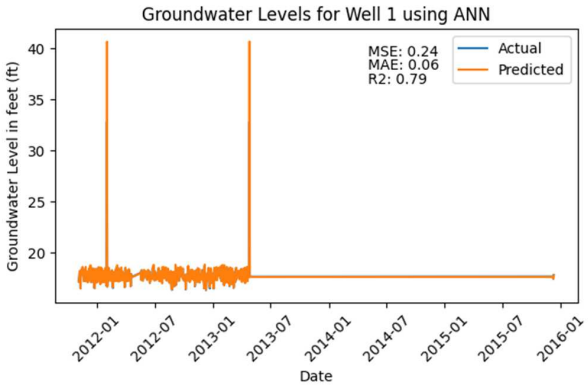


Fig. 6. Comparison of the actual and predicted groundwater levels (testing set) in the first well using ANN

B. *LSTM model*

The LSTM model's training and testing results for forecasting groundwater levels in wells are represented as follow:

In the training set, the first well, the LSTM model achieves a low MSE near zero, a MAE of 0.03, and a strong R2 of 0.99, indicating an excellent fit to the training data. Similarly, third well and fourth well 4 both exhibit robust training set performance, with low MSE values of 0.00, MAE values of 0.02 and 0.05, and high R2 values of 0.99 and 0.98, respectively.

For well one in the testing set, the LSTM model shows a MAE of 0.1498, a RMSE of 1.2200, and an R2 of 0.2663, indicating a moderate fit but with some room for

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improvement. While second Well maintains its impressive performance in the testing set, with a very low MAE of 0.0073, an exceptionally low RMSE of 0.0099, and a high R2 of 0.9981. Well three also performs well in the testing set, showing a low MAE of 0.0178, a low RMSE of 0.0225, and a high R2 of 0.9959, and for well four, the testing set results indicate a moderate fit with an MAE of 0.0326, an RMSE of 0.0407, and an R2 of 0.9717.

These results highlight the ability of the LSTM model to capture complex temporal patterns in groundwater data. While the model outperformed training, there were some differences in the performance of the testing sets, with Well 1(Fig. 7) showing the greatest potential for improvement. Further data preprocessing and model optimization techniques can improve its prediction accuracy, especially in difficult cases like Well 1. Overall, the LSTM model shows promise as a valuable tool for predicting groundwater levels for a variety of wells.

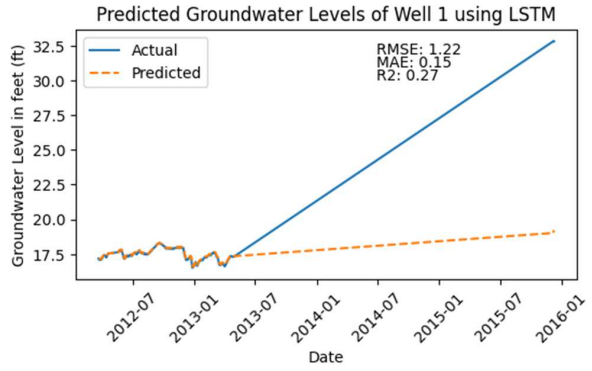


Fig. 7. Comparison of the actual and predicted groundwater levels (testing set) in the first well with LSTM

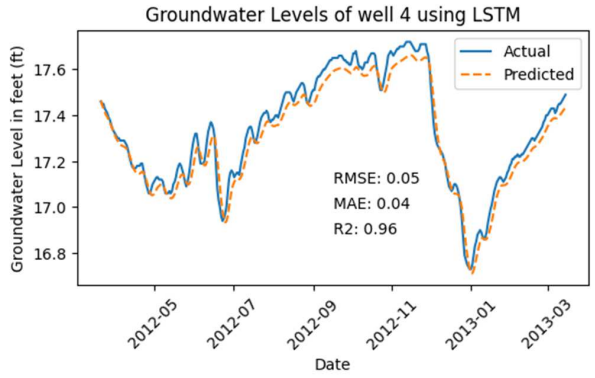


Fig. 8. Comparison of the actual and predicted groundwater levels (testing set) in the fourth well with LSTM

C. *XGBoost model*

The modeling results obtained by using XGBoost model for the validation and the training data are represented as follow:

For well one in the validation set, the XGBoost model shows a MAE of 0.1766, a MSE of 0.1136, a RMSE of 0.3371, and an R2 score of 0.8629. These measures indicate a reasonable fit to the validation data, with the R2 score showing that the model explains a significant portion of the variance, while Well 3 shows excellent performance in the validation set, with an extremely low MAE of 0.0073, an MSE of 9.62e-05, a very low RMSE of 0.0098, and an almost perfect R2 score of 0.9992. These results indicate that the XGBoost model is particularly suitable for validation data, effectively capturing underlying patterns.

For well one of the test set, the XGBoost model yielded an RMSE of 0.3190, an MAE of 0.1675, and an R2 score of 0.8795. These results show a good fit to the testing data, with the R2 score suggesting that the model explains a significant portion of the variance (Fig.9). Well three maintained outstanding performance across the test set, with a very low RMSE of 0.0093, an extremely low MAE of 0.0070, and a near-perfect R2 score of 0.9993. These results indicate that the XGBoost model generalizes extremely well to unseen data, reaffirming its predictive ability (Fig.10).

These results highlight the effectiveness of the XGBoost model in predicting groundwater level. Although Well 1 showed slightly more variability in test results, the model showed strong overall performance. The exceptionally high R2 scores for Well 3 in both the validation and test sets highlight the model`s ability to capture complex relationships in the data. Additional enhancements and optimizations could improve model accuracy, especially for well one, but overall, XGBoost proves to be a valuable tool for groundwater forecasting for these wells.

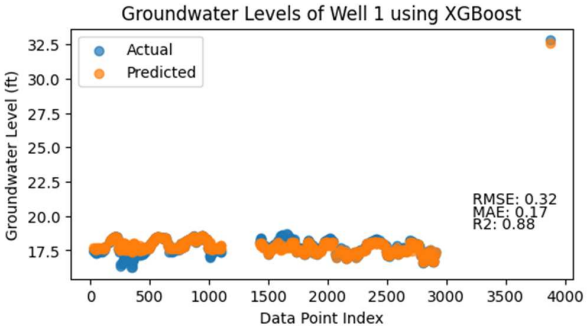


Fig. 9. Comparison of the actual and predicted groundwater levels (testing set) in the first well with XGBoost

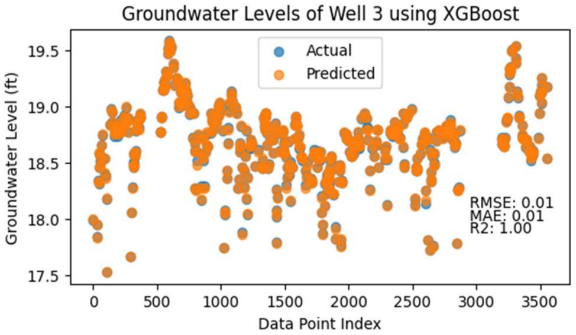


Fig. 10. Comparison of the actual and predicted groundwater levels (testing set) in well three with XGBoost

D. *Performance comparison*

The results showed that the FFNN, LSTM and the XGBoost models performed generally well for forecasting of groundwater level of the four wells, with only marginal differences. The results also indicate that the Deep Learning

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model LSTM did not show significant superiority over the traditional FFNN and the decision tree XGBoost model.

This observation was similar to that reported in the study by Malek and Nouiri [21] and the study of Suja and Dr. Sindhu [22] which adopted ANN based models to predict groundwater level (GWL), the results indicate the efficiency of ANN with small range error. However, B. S. Supreetha [23] and J.Zhang and Y.Zhu [24] concludes that GWL forecasting with LSTM performs much better than FFNN especially with systematically configured LSTM models. In term of XGBoost, A.Osman and A.Najah [25] found that XGBoost model presented more accurate prediction results with the highest R2, and smaller MAE, and RMSE values the ANN models.

The difference in the results is showed in Fig.11 with the performance measure R-Squared R2 and Mean absolute error MAE Fig.12 as follow.

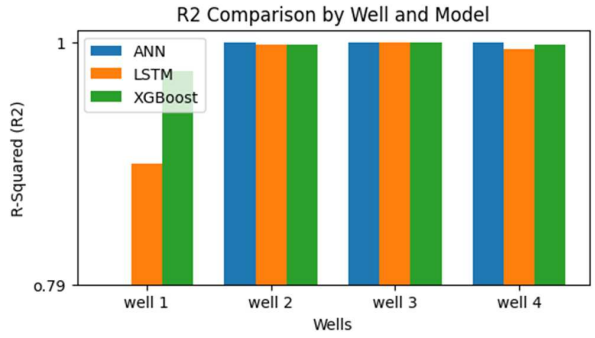


Fig. 11. Comparison of the model performance between FFNN, LSTM and XGBoost using the R-Squared measure

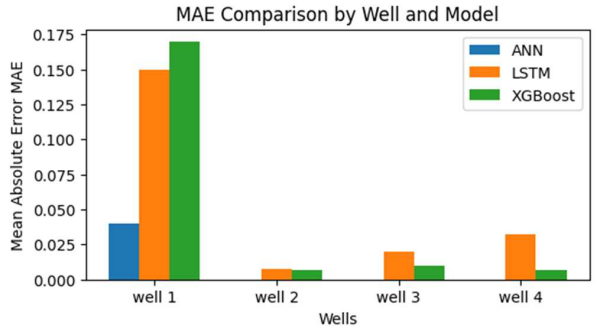


Fig. 12. Comparison of the model performance between FFNN, LSTM and XGBoost using the mean absolute error measure

VI. CONCLUSION

The application of machine learning models for forecasting groundwater levels in multiple wells, as shown in this work for case of the Marrakech area, represents a significant advancement in the field of hydrogeology and environmental monitoring. This study has not only showcased the potential of data-driven approaches in addressing the complex challenges of groundwater management but has also highlighted the importance of leveraging modern technology to better understand and predict groundwater dynamics.

The results of this research have shown that machine learning models, when trained on historical data and equipped with the right features, can provide accurate and reliable forecasts of groundwater levels. This information can be invaluable for water resource management, agriculture, urban planning, and various other sectors that rely on a sustainable and well-managed groundwater supply.

The successful implementation in this study underscores the importance of collaboration between hydrogeologists, data scientists, and policymakers to harness the full potential of such tools. As we move forward, it is imperative to continue refining these models, incorporating real-time data, and expanding their applicability to other regions facing similar groundwater challenges.

As we look to the future, this research serves as a stepping stone for further advancements in hydrogeology, ultimately contributing to more informed decision-making and the preservation of groundwater resources in the Marrakech area and beyond.

VII. REFERENCES

[1] Qadir, M., Sharma, B. R., Bruggeman, A., Choukr-Allah, R., & Karajeh, F. (2007). Non-conventional water resources and opportunities for water augmentation to achieve food security in water scarce countries. Agricultural Water Management, 87(1), 2–22. https://doi.org/10.1016/j.agwat.2006.03.018

[2] Wada, Y., Van Beek, L. P. H., Van Kempen, C. M., Reckman, J. W., Vasak, S., & Bierkens, M. F. P. (2010). Global depletion of groundwater resources. Geophysical Research Letters, 37(20), n/a. https://doi.org/10.1029/2010gl044571

[3] Malik, A., & Bhagwat, A. (2021). Modelling groundwater level fluctuations in urban areas using artificial neural network. Groundwater for Sustainable Development, 12, 100484. https://doi.org/10.1016/j.gsd.2020.100484

[4] Rajaee, T., Ebrahimi, H., & Nourani, V. (2019). A review of the artificial intelligence methods in groundwater level modeling. Journal of Hydrology, 572, 336–351. https://doi.org/10.1016/j.jhydrol.2018.12.037

[5] S.S. Nair, G. Sindhu, Groundwater level forecasting using artificial neural network, Int. J. Sci. Res. Publ. 6 (1) (2016) 2250–3153.

[6] S. Lee, K.-K. Lee, H. Yoon, Using artificial neural network models for Groundwater level forecasting and assessment of the relative impacts of Influencing factors, Hydrogeol. J. 27 (2) (2019) 567–579.

[7] Kayhomayoon, Z., Babaeian, F., Milan, S. G., Azar, N. A., & Berndtsson, R. (2022). A combination of metaheuristic optimization algorithms and machine learning methods improves the prediction of groundwater level. Water, 14(5), 751. https://doi.org/10.3390/w14050751

[8] Roy, D. K., Biswas, S. K., Mattar, M. A., El-Shafei, A. A., Murad, K. F. I., Saha, K. K., Datta, B., & Dewidar, A. Z. (2021). Groundwater level prediction using a multiple objective Genetic Algorithm-Grey relational analysis based weighted ensemble of ANFIS models. Water, 13(21), 3130. https://doi.org/10.3390/w13213130

[9] Adnan, R. M., Dai, H. L., Mostafa, R. R., Islam, A. R. M. T., Kisi, O., Heddam, S., & Zounemat-Kermani, M. (2022). Modelling groundwater level fluctuations by ELM merged advanced metaheuristic algorithms using hydroclimatic data. Geocarto

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International, 38(1). https://doi.org/10.1080/10106049.2022.2158951

[10] Hawari, H. F., & Chantar, P. P. (2021). Development of IoT Real-Time Groundwater Monitoring System. Journal of Physics, 2107(1), 012032. https://doi.org/10.1088/1742-6596/2107/1/012032

[11] Jaisudhan, A. J. P. (2023). IoT based Ground Water Level Monitoring System and Its Impact in Agricultural Applications. Research Square (Research Square). https://doi.org/10.21203/rs.3.rs-2525548/v1

[12] Aderemi, B. A., Olwal, T. O., Ndambuki, J. M., & Rwanga, S. S. (2021). A Review of Groundwater Management Models with a Focus on IoT-Based Systems. Sustainability, 14(1), 148. https://doi.org/10.3390/su14010148

[13] Tao, H., Hameed, M. M., Marhoon, H. A., Zounemat-Kermani, M., Heddam, S., Kim, S., Sulaiman, S. O., Tan, M. L., Sa'adi, Z., Mehr, A. D., Allawi, M. F., Abba, S., Zain, J. M., Falah, M. W., Jamei, M., Bokde, N. D., Bayatvarkeshi, M., Al-Mukhtar, M., Bhagat, S. K., . . . Yaseen, Z. M. (2022). Groundwater level prediction using machine learning models: A comprehensive review. Neurocomputing, 489, 271–308. https://doi.org/10.1016/j.neucom.2022.03.014

[14] Hussein, E. A., Thron, C., Ghaziasgar, M., Bagula, A., & Vaccari, M. (2020). Groundwater Prediction using Machine-Learning Tools. Algorithms, 13(11), 300. https://doi.org/10.3390/a13110300

[15] Aldhyani, T. H. H., Al-Yaari, M., Alkahtani, H., & Maashi, M. (2020). Water quality prediction using artificial intelligence algorithms. Applied Bionics and Biomechanics, 2020, 1–12. https://doi.org/10.1155/2020/6659314

[16] Zhu, S., Hrnjica, B., Ptak, M., Choiński, A., & Sivakumar, B. (2020). Forecasting of water level in multiple temperate lakes using machine learning models. Journal of Hydrology, 585, 124819. https://doi.org/10.1016/j.jhydrol.2020.124819

[17] Wünsch, A., Liesch, T., & Broda, S. (2021). Groundwater level forecasting with artificial neural networks: a comparison of long short-term memory (LSTM), convolutional neural networks (CNNs), and non-linear autoregressive networks with exogenous input (NARX). Hydrology and Earth System Sciences, 25(3), 1671–1687. https://doi.org/10.5194/hess-25-1671-2021

[18] Osman, A. I. A., Ahmed, A. N., Fai, C. M., Huang, Y. F., & El-Shafie, A. (2021b). Extreme gradient boosting (Xgboost) model to predict the groundwater levels in Selangor Malaysia. Ain Shams Engineering Journal, 12(2), 1545–1556. https://doi.org/10.1016/j.asej.2020.11.011

[19] Kim, D., Jang, C., Choi, J., & Kwak, J. (2023). A case study: Groundwater level forecasting of the Gyorae Area in actual practice on Jeju Island using Deep-Learning technique. Water, 15(5), 972. https://doi.org/10.3390/w15050972

[20] C. Chen, X. Zhu, X. Kang and H. Zhou, "A Deep Learning Algorithm for Groundwater Level Prediction based on Spatial-temporal Attention Mechanism," 2021 IEEE Intl Conf on Dependable, Autonomic and Secure Computing, Intl Conf on Pervasive Intelligence and Computing, Intl Conf on Cloud and Big Data Computing, Intl Conf on Cyber Science and Technology Congress (DASC/PiCom/CBDCom/CyberSciTech), AB, Canada, 2021, pp. 716-723, doi: 10.1109/DASC-PICom-CBDCom-CyberSciTech52372.2021.00119

[21] Derbela, M., Nouiri, I. Intelligent approach to predict future groundwater level based on artificial neural networks (ANN). Euro-Mediterr J Environ Integr 5, 51 (2020).

[22] Suja S Nair, Dr. Sindhu G (2018); Groundwater level forecasting using Artificial Neural Network; Int J Sci Res Publ 6(1) (ISSN: 2250-3153). http://www.ijsrp.org/research-paper-0116.php?rp=P494995

[23] Supreetha, B. S., Shenoy, N. K., & Nayak, P. L. (2020). Lion Algorithm-Optimized Long Short-Term Memory Network for groundwater level forecasting in Udupi District, India. Applied Computational Intelligence and Soft Computing, 2020, 1–8. https://doi.org/10.1155/2020/8685724

[24] Zhang, J., Zhu, Y., Zhang, X., Ye, M., & Yang, J. (2018). Developing a Long Short-Term Memory (LSTM) based model for predicting water table depth in agricultural areas. Journal of Hydrology, 561, 918–929. https://doi.org/10.1016/j.jhydrol.2018.04.065

[25] Osman, A. I. A., Ahmed, A. N., Fai, C. M., Huang, Y. F., & El-Shafie, A. (2021). Extreme gradient boosting (Xgboost) model to predict the groundwater levels in Selangor Malaysia. Ain Shams Engineering Journal, 12(2), 1545–1556. https://doi.org/10.1016/j.asej.2020.11.011

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